R02 native factorization continuation

Run nist-factor-R02-20261022T0724Z; production began 2026-10-22 07:24 UTC. The frozen base protocol, its additive NFS protocols, per-attempt source commits and receipts define this round. Baseline is R01 at crypto 99132d1fd1a9d9c125eec4c1a9761edce068076a.

Requested work and verified outcome

The user requested continued factorization of the skipped-candidate DiSSECT results. All three unresolved lineages received native searches. All nine partitions were recertified and replayed in separate PARI processes against targets reconstructed from the original hexadecimal curve fixture. Eight targets are complete, one partial; thirteen of the fourteen original timeout-valued fields can now be replaced. The earlier upstream suite contains 438 cases and ten timeout rows; this round is a factorization supplement, not a fresh execution of that suite.

Requirement	Status and evidence
Continue all three unresolved factorizations	Executed with the frozen allocations; raw targets/ and nfs/
Prime-certify every claimed factor	Verified; loose certificates/final/*/certs-final/
Reconstruct inputs and independently replay products	Verified for all nine; raw validation/final-loose-replay/
Complete every factorization	Partial: only P-256 block 876 remains composite
Product identities and identity records	Verified: nine Lean theorems, nine separately checked records
Preserve and publish all raw evidence	Preserved locally; large-archive S3 publication blocked by missing AWS identity
Explain historical skips	Not established by these computations

Recorded arithmetic results

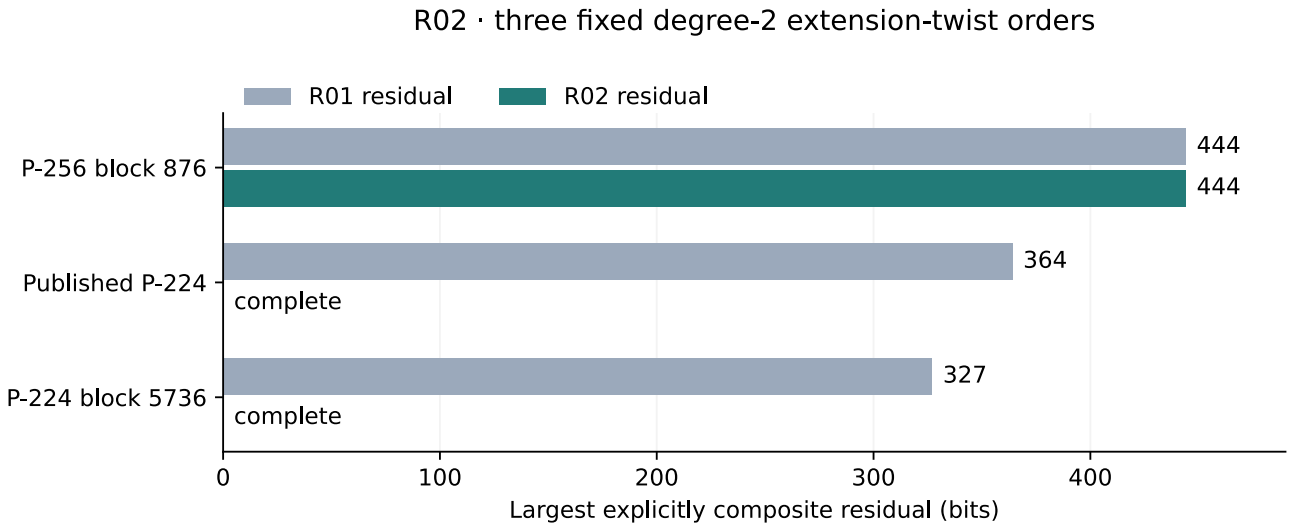


Figure: largest explicitly composite residual, in bits, before and after R02, on the three fixed lineages (one integer each). Zero denotes complete prime factorization. Source: native verifier fields in R01 baseline and R02 results, transcribed by plot-data.json; no sampling uncertainty. No running-time comparison is plotted.

The 327-bit residual for P-224 block 5736 split on ECM curve 29 into the 40- and 59-digit primes printed in the round README. Its complete 448-bit order has largest prime of 196 bits. The published P-224 364-bit residual split by resumed NFS into 47- and 63-digit primes; the complete 448-bit order has largest prime of 209 bits. Exact decimal partitions and exponents are in results.json.

The only residual is the 444-bit composite


```
33492266050798185653682380062483799213318454801653963493134064721059735936944520907295931496614455858076048437184499090478420204888353
```


It belongs to the 512-bit degree-2 extension-twist order of skipped_256_876; its largest currently proven prime factor is 33 bits. This says nothing about the smallest prime factor of the residual.

Methods, controls and failures

Native PARI perfect-power and cross-panel GCD checks found no proper split. Read-only FactorDB responses reported all three composites without new factors. An initial asynchronous lookup failed to await its pipelines; both that failed attempt and the successful hints-v2/ lookup are retained.

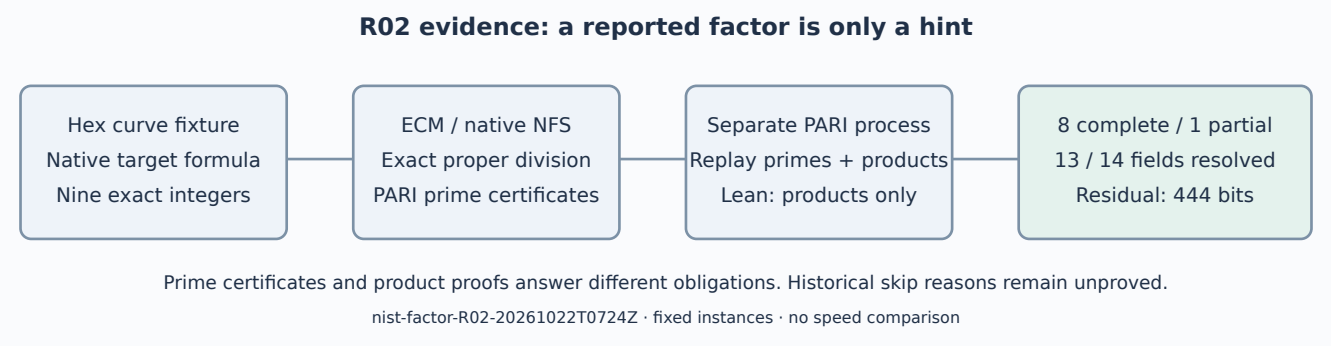
Each lineage received one base-2 P−1 run at B1=10,000,000, B2=10,000,000,000; eight P+1 starts 6,7,11,13,17,19,23,29 at B1=1,000,000, B2=1,000,000,000; and up to 100 ECM curves at B1=11,000,000, native B2=35,133,391,030. P−1/P+1 yielded no split. P-224 block 5736 attempted/completed 29 ECM curves, exit 14 with a split; published P-224 completed 100, exit 0; P-256 block 876 attempted 92 and completed stage 2 for 91, exit 124 at the process limit. Unused allocations are respectively 71, 0 and 8; none is reported as executed. Positive P−1 and P+1 controls yielded the expected exact proper divisors.

The two up-to-600-second factorint(C,6) phases emitted a too-large-MPQS warning and then a matrix-index serialization error after the alarm. No usable proposal survived. These are inconclusive implementation failures, not valid negative factorization results. The verified previous partitions were recovered unchanged. The corrected matrix guard passed separate small-matrix and one-second alarm controls, without rerunning production or resetting its budget. A proper new hint passed native refinement; an invalid divisor 2 for the odd input was rejected.

YAFU was built from pinned source c7d90283972820f99b3b4920bfc1eff05c1f6342, AVX2/BMI2 and GMP-ECM enabled, with matching Debian libecm headers. Missing-header build errors, the initial stdin/seed setup errors and a checksum glob error are retained. The corrected 85-digit known-answer control first auto-selected SIQS. It is not GNFS validation. A separate -gnfs -xover 81 control collected its relations within a 300-second allocation, timed out during filtering, then completed and verified both expected prime factors in a separately frozen 300-second resume. See raw controls/nfs-gnfs-positive/ and controls/nfs-verification/result.json.

Production NFS used two threads, two linear-algebra threads in their separate phase, seed 20261022, native limit 1800 seconds, outer limit 1860 seconds and 30-second kill grace, per-process virtual memory limit 4 GiB. The published P-224 attempt exited 139 after retaining about 6.94 million relations; the cause is unknown. Its separately frozen resume retained 10,830,564 relations, filtered, solved the matrix, and produced the two certified factors, exit 0. Filtering logged relation warnings; final prime and exact-product checks passed independently. P-256 terminated with exit 137 in polynomial search; only search state survived. No selected polynomial, sieve completion or valid negative arithmetic result is claimed. An event-based hardlink helper retained native outputs across cleanup; single-/multiple-owner create-write-unlink smoke controls passed.

Proofs and validation



The proof map links all nine integer equalities and identity.certificate/v1 records. Lean 4.35.0 proof gate: nine theorems, zero failing, zero trusted native proofs, empty axiom lists. The Lean proofs establish products over Nat; PARI certificates establish primality. lean4checker was unavailable. The native repository identity checker issued and separately checked all nine records with 32 extra modular evaluation points. Modular identity checking alone is not the integer proof.

Before the main-branch merge, 4,474 library tests passed; after it, 4,479 passed, zero failed and 119 were ignored. Network access was granted for the loopback-dependent tests. No new library implementation was benchmarked. The host has four CPUs and 16 GiB RAM; this unisolated session is L0. Setup, failed builds, controls and production searches are preserved, but their operational timings do not support a speed comparison. Native tools: PARI/GP 2.17.2, GMP-ECM 7.0.6; executable/siever hashes are recorded in raw receipts. Python is used only for plotting recorded values and PDF layout.

Evidence and open work

The archive manifest binds the complete local raw archive. It preserves successes, errors, timeouts, terminated searches, native checkpoints, source snapshots and scoped Git commit patches. An optional full Git bundle failed because the blobless clone required unavailable promisor objects; that failure is retained too. The runtime has no configured AWS identity, so publication through the required S3 artifact store remains an access blocker; no remote object is claimed. Loose mathematical certificates and a small P-256 checkpoint excerpt make verification and resumption reviewable without the bulk archive.

The specific obstruction requires a complete, prime-certified split of the remaining P-256 cofactor. R03 found that the original per-thread polynomial excerpt restarts search without a master file. Its exact validation and native-recognized recovery are separate from R02. Use that corrected checkpoint for continuation; the wrapper now refuses the original per-thread-only state. R03 must not be counted as an R02 attempt. The original missing generators and upstream torsion/trace defects remain outside this supplement. A finite search cannot establish a historical selection motive or curve security.

The new residual-bit graph and proof diagram are local to this round. IC scoreboards/timeline, curve diagrams, performance-gains figures and seed-study visuals contain no affected factorization claim and remain unchanged. Conductor was unavailable; the documented fallback was used.

Sources: upstream DiSSECT, pinned YAFU, Anthropic origin story, Valsorda seed report.